

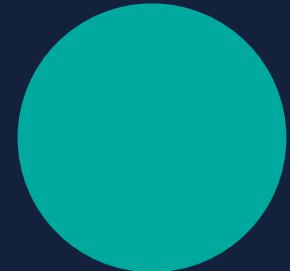
AI ENGINEER CANDIDATE ASSIGNMENT

AI-Powered Credit Risk Intelligence Platform

An end-to-end, explainable ML platform for loan default risk assessment, built on the Home Credit Default Risk dataset.

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Business Problem & Objective



The Challenge

- Banks face pressure to make faster, more accurate, and explainable credit decisions.
- High-risk applicants often are not identified early enough in manual review workflows.
- Regulators and auditors increasingly expect risk decisions to be explainable, not opaque.
- Business teams often lack a fast way to explore applicant data without writing SQL.



Platform Objective

- Predict the probability that a loan applicant will default.
- Convert probability into LOW / MEDIUM / HIGH risk bands.
- Compare multiple ML models and select the strongest performer.
- Let users explore applicant data through natural language (Talk-to-Data).
- Persist applicant data and prediction logs in SQLite.
- Deliver everything through an interactive Streamlit UI, deployed via Docker.



Assignment Requirements & Scope



EDA

Business insights & data quality analysis



Machine Learning

Default probability prediction model



Explainability

Understandable prediction reasoning



Talk-to-Data

Conversational natural-language querying



Business Rules

Business-readable decision insights



User Interface

Multi-section Streamlit application



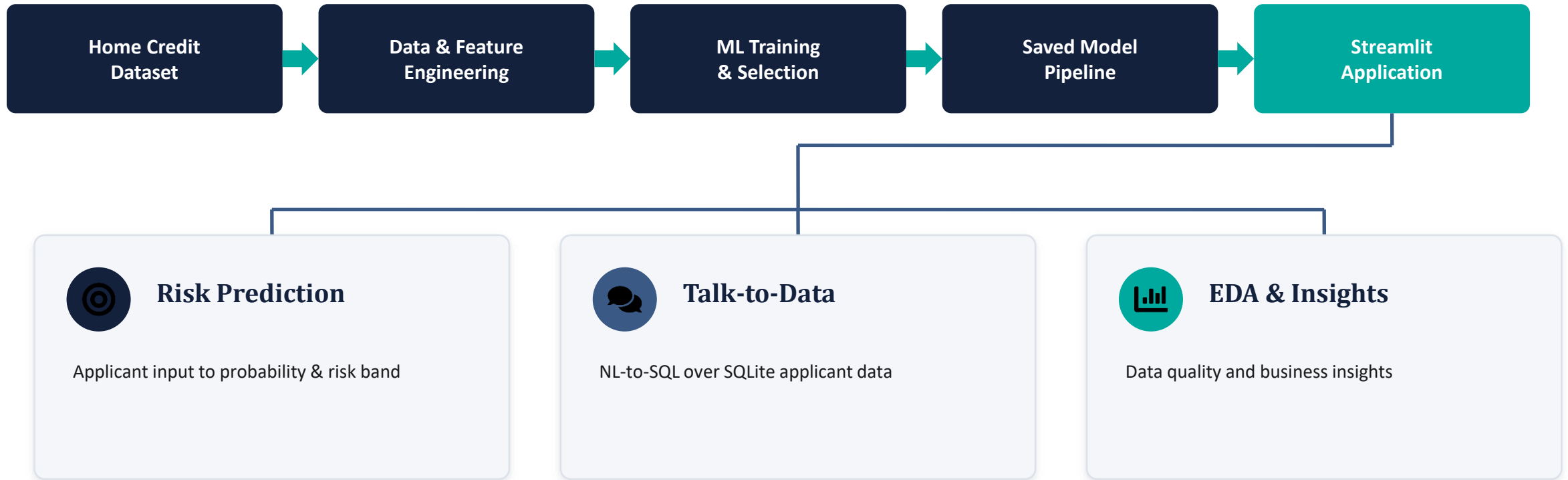
Dockerization

Reproducible containerized deployment

Evaluation weighting: ML Design 30% | LLM / Talk-to-Data 25% | EDA 15% | Prompt & Hallucination Control 15% | Dockerization 10% | Documentation 5%



Solution Overview



One centralized feature-engineering layer and one risk-classification function are shared across training, prediction, and analysis - preventing drift between components.



Dataset & Data Understanding

Home Credit Default Risk Dataset

300,000+

Original applicant records

122

Original raw columns

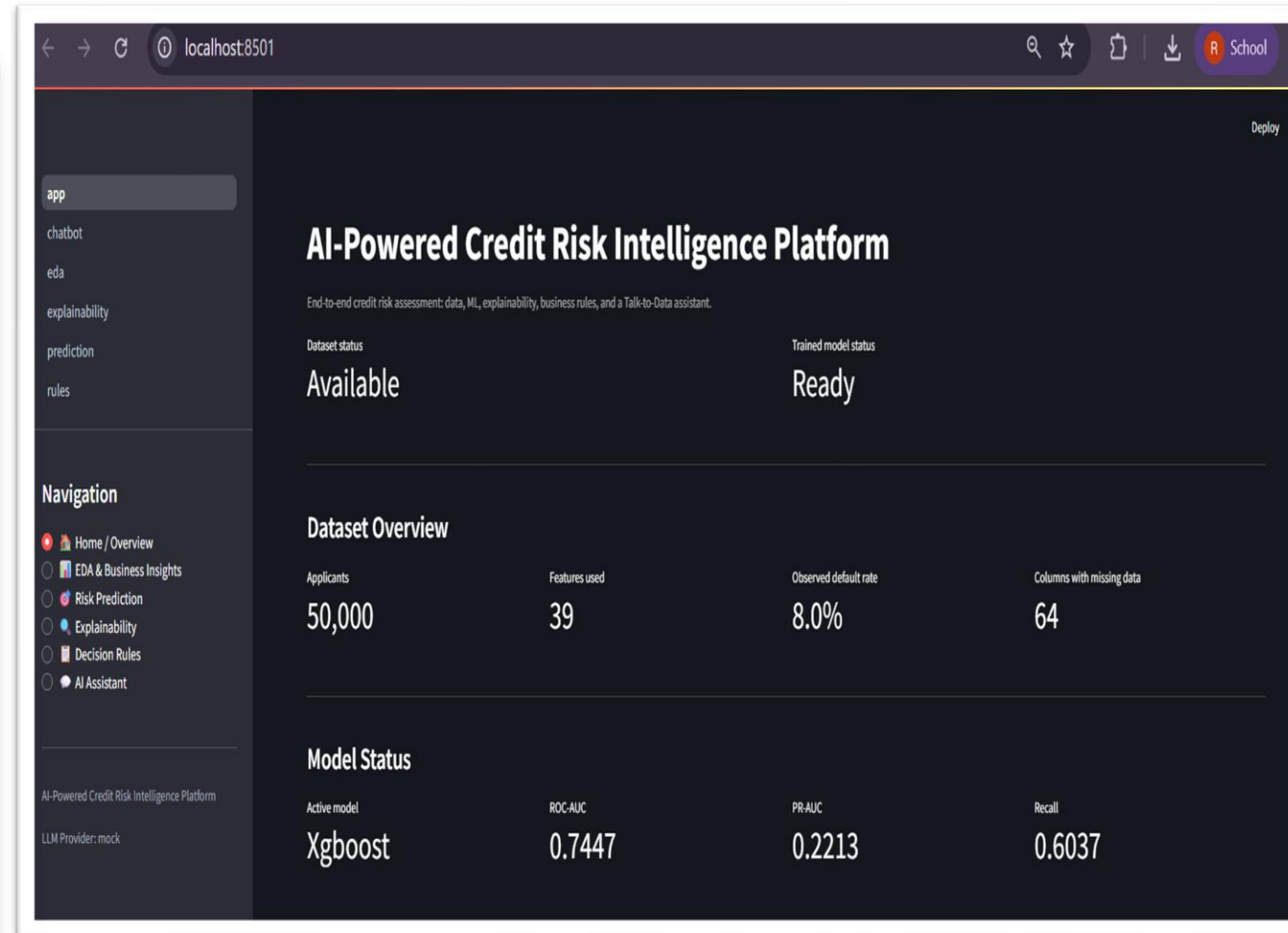
TARGET

0 = No Default | 1 = Default



Engineering Note: Memory-Aware Sampling

Loading the full dataset on the local development machine hit memory limits. The pipeline supports a configurable DATA_SAMPLE_SIZE; a 50,000-row sample enabled successful local training and testing. Full-dataset training is supported in a higher-memory environment.





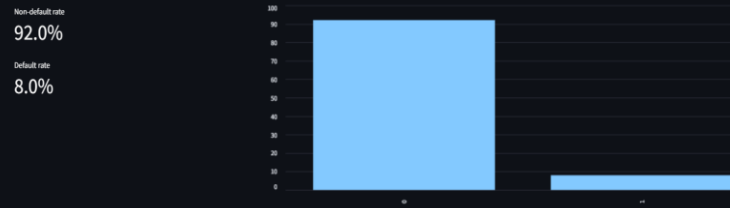
EDA & Key Insights

At least 5 business insights are surfaced directly from the data. Insert the exact insight titles, figures, and supporting charts from your EDA run below.

1

Target (Default) Distribution

Business Insight 1 — Target (Default) Distribution



Business interpretation: the dataset is heavily imbalanced — most applicants repay their loans. This is why the ML layer evaluates with ROC-AUC/PK-AUC and class-weighting rather than plain accuracy.

2

External Credit Scores vs. Default

Business Insight 2 — External Credit Scores vs. Default



Business interpretation: applicants who defaulted (TARGET=1) tend to have lower average external credit scores than those who repaid — consistent with these scores being pre-existing risk indicators.

3

Default Rate by Income Type

Business Insight 3 — Default Rate by Income Type



Business interpretation: default rates vary meaningfully across income types, from 0.0% to 100.0% — income type alone is not the whole story, but it's a useful segmentation for policy discussions.

4

Credit-to-Income Ratio vs. Default

Business Insight 4 — Credit-to-Income Ratio vs. Default



Business interpretation: a higher credit-to-income ratio means the requested loan is large relative to the applicant's income — worth comparing between defaulters and non-defaulters as a leverage indicator.

5

Missing Value Patterns

Business Insight 5 — Missing Value Patterns



Business interpretation: columns with high missingness (e.g. external scores or employment fields) often reflect applicants with thinner credit files — missingness itself can carry risk signal, not just noise.



Data Processing & Feature Engineering

Centralized Dataset Loader

- ✓ Dataset file existence & structural validation
- ✓ Required-column validation before use
- ✓ Memory-efficient, configurable-sample loading
- ✓ Configurable sampling via DATA_SAMPLE_SIZE
- ✓ Reproducibility enforced through RANDOM_SEED

39

Modeling Features

27

Numerical

12

Categorical

Feature engineering and cleaning logic is defined once and reused identically for training and inference.



Machine Learning Approach



Logistic Regression

Interpretable linear baseline



Random Forest

Bagged ensemble of decision trees



XGBoost

Gradient-boosted ensemble



Class Imbalance

In the 50,000-row training sample, only ~8.05% of applicants belong to the positive (default) class.

A model predicting "no default" for everyone would score deceptively well on accuracy - accuracy is therefore not used for model selection.



Why PR-AUC

PR-AUC focuses on the precision-recall trade-off for the minority (default) class - exactly the class this platform must identify well.

Selection order: PR-AUC then Recall (tie-break) then Inference speed (final tie-break).



Model Comparison & Results

Model	ROC-AUC	PR-AUC	Precision	Recall	F1-Score	Inference (ms/row)
Logistic Regression	0.7479	0.2279	0.1605	0.6832	0.2600	0.101
Random Forest	0.7446	0.2124	0.2015	0.5205	0.2906	0.9611
XGBoost (Selected)	0.7460	0.2318	0.1898	0.6075	0.2893	0.1889



XGBoost Selected

Ranked #1 on PR-AUC (0.2318) - the primary, imbalance-aware selection metric - ahead of Logistic Regression (0.2279) and Random Forest (0.2124).

Selection strategy: rank by PR-AUC, tie-break on minority-class recall, then on inference speed. Accuracy is intentionally excluded given the class imbalance.



Credit Risk Prediction



LOW RISK

Probability < 0.30

MEDIUM RISK

0.30 <= Probability < 0.60

HIGH RISK

Probability >= 0.60

Credit Risk Prediction

Enter applicant details to get a default-risk assessment.

Demographics	Financial	Employment & Credit Signal
Gender F	Annual income 90000	Income type Working
Number of children 0	Requested credit amount 600000	Occupation Laborers
Family members 2	Loan annuity 27000	Years employed (0 if not employed) 5
Family status Married	Goods price 540000	External credit score #1 0.50
Age (years) 35	Owns a car N	External credit score #2 0.50
	Owns real estate N	External credit score #3 0.50

Assess Risk

Result

MEDIUM RISK — 49.4% default probability

See the Explainability and Rules pages for why this applicant received this classification.



Talk-to-Data System



Mock Provider (Default)

Deterministic, offline, zero API cost. Covers all required query patterns for development, testing, and demonstration.



Anthropic Provider (Optional)

Enabled via LLM_PROVIDER, ANTHROPIC_API_KEY, ANTHROPIC_MODEL - no code changes required to switch.

The screenshot displays the user interface of the Talk-to-Data System. It shows a chat history with four previous queries and their corresponding SQL results. The first query asks for the minimum number of children, the second for the average number, the third for the number of occupation groups, and the fourth for the occupation groups with the highest default rate. The fifth query, which is the current one, asks for the occupation groups by observed default rate, and the interface shows a list of five groups: Low-skill Laborers (15.57%), Waiters/barmen staff (11.84%), IT staff (11.54%), Drivers (11.15%), and Security staff (10.69%). The highest rate is identified as 15.57% for Low-skill Laborers. Each query and result is followed by a 'Generated SQL' button.

Chat History:

- Query 1:** The minimum number of children recorded for an applicant is 0.
Generated SQL
- Query 2:** What is the average number of children?
Response: The average number of children per applicant is 0.42.
Generated SQL
- Query 3:** How many occupation groups are there?
Response: There are 18 distinct occupation groups in the dataset.
Generated SQL
- Query 4:** Which occupation groups have the highest default rate?

Current Query: Occupation groups by observed default rate

Results:

1. Low-skill Laborers — 15.57%
2. Waiters/barmen staff — 11.84%
3. IT staff — 11.54%
4. Drivers — 11.15%
5. Security staff — 10.69%

Summary: The highest observed default rate in this ranking is for Low-skill Laborers, at 15.57%.
Generated SQL



Prompt Optimization & Hallucination Control



Schema-Aware Querying

The model only sees the actual applicant table schema - not the full database.



SQL Validation Before Execution

Every generated query is parsed and validated before it ever touches SQLite.



Read-Only Execution

Only SELECT statements are permitted; write/DDL operations are rejected outright.



Restricted, Business-Relevant Schema

Only a curated, business-relevant column subset is exposed to the chatbot.



Row Limits on Results

Returned rows are capped before being passed further down the pipeline.



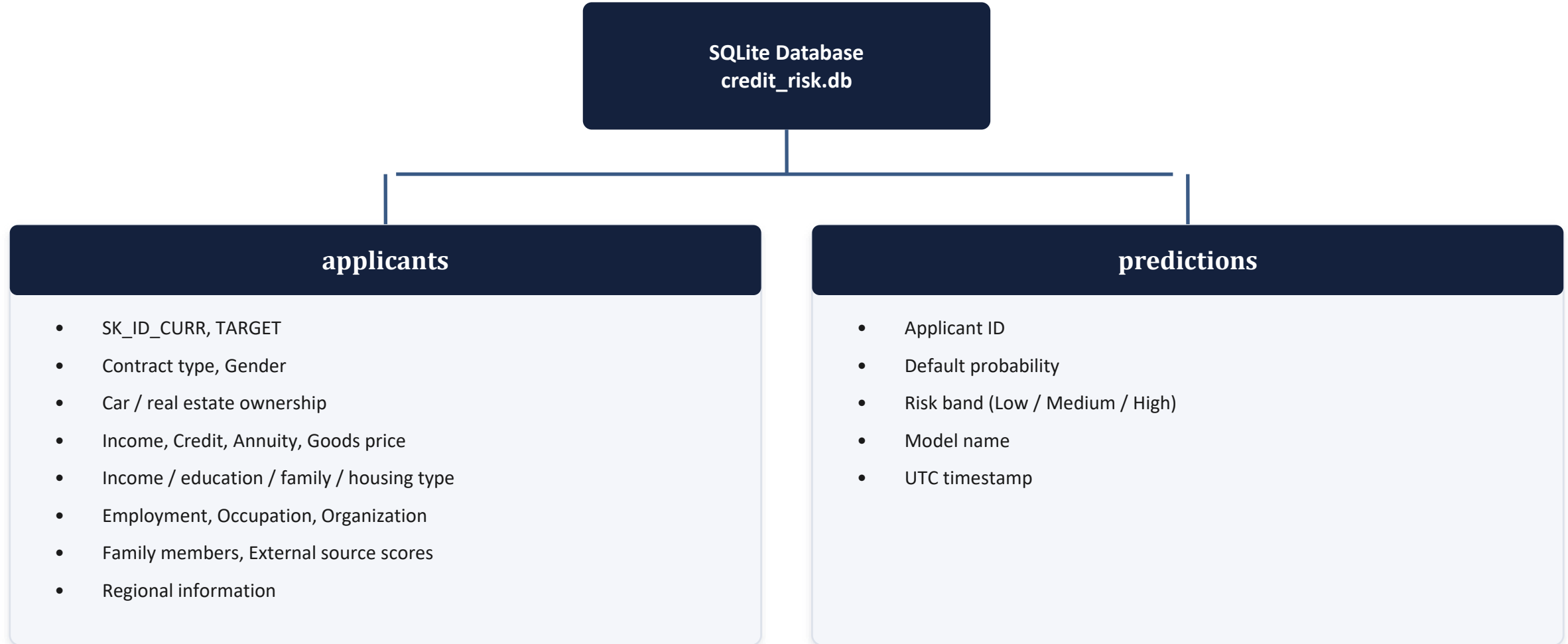
Conversation Memory (3 Turns)

CONVERSATION_MEMORY_TURNS=3 bounds context - no unbounded history growth.

Deterministic mock provider keeps supported query patterns fully predictable - no hallucinated SQL is possible for patterns it explicitly implements.



Database & System Design





Explainability & Business Understanding

Current Explainability Approach

- SHAP (TreeExplainer / LinearExplainer) computes global feature importance - mean absolute SHAP value - across a sample of applicants.
- Local, per-applicant SHAP values identify the top factors increasing and reducing that specific prediction.
- Technical feature names are translated into business-readable labels (e.g. EXT_SOURCE_2 -> "External credit score #2").
- If SHAP computation fails for any reason, the interface shows a clean fallback rather than a fabricated explanation.

Prediction Explainability

Understand which factors influence the model globally and why it produced the latest applicant's risk prediction.

MEDIUM RISK — 49.4% default probability

Probability of Default
49.4%

Model confidence (distance from the 50% decision threshold): 1%. This measures how far the prediction is from the point of maximum uncertainty and is not a calibrated statistical confidence interval.

The model estimates a 49.4% probability of default, placing this applicant in the MEDIUM RISK category.

Local Explanation – Last Prediction

This applicant was classified as MEDIUM RISK with a 49.4% predicted probability of default.

Risk Contribution Breakdown

Each bar shows how strongly an individual feature pushes this specific prediction toward lower or higher predicted default risk.





Dockerized Deployment

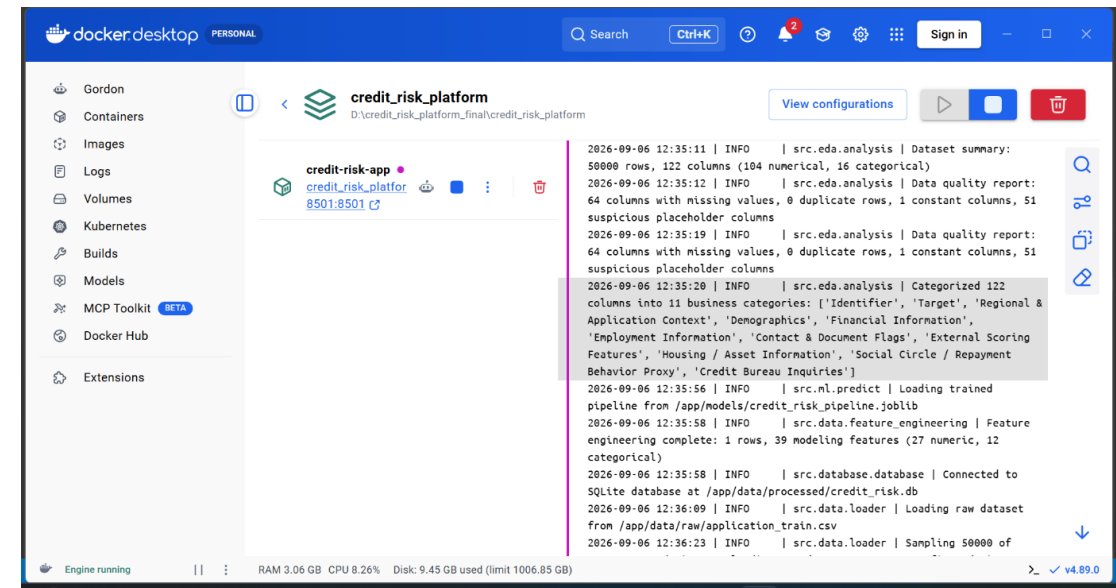
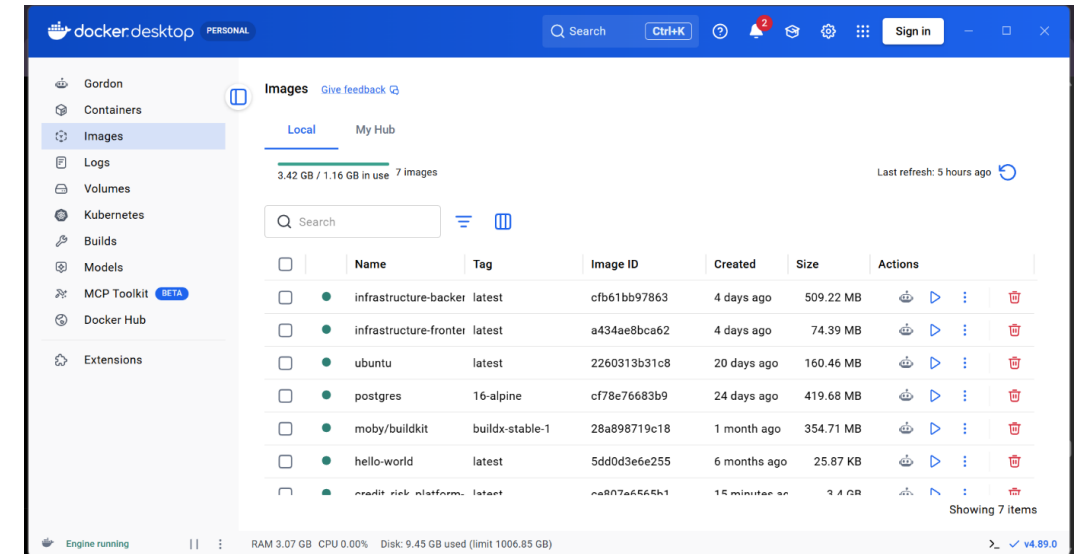
Configuration

- Python 3.11-slim base image
- Dependencies installed, project copied, port 8501 exposed
- Built-in Streamlit health check
- Compose: port 8501:8501, .env loaded
- Volumes: ./data and ./models mounted from host
- restart: unless-stopped

docker compose up --build

Container Starts

App Available on localhost:8501





Engineering Challenges & Solutions

Challenge	Solution
Memory limits loading the full 300,000+ row dataset	Configurable DATA_SAMPLE_SIZE; successful 50,000-row local training run
Heavy class imbalance (~8% positive class)	class_weight / scale_pos_weight plus PR-AUC-driven model selection
Making applicant data accessible to non-technical users	Talk-to-Data natural-language querying interface
Safe natural-language-to-SQL querying	Restricted schema, SQL validation, and read-only execution
Reproducible, portable deployment	Dockerfile + docker-compose with host-mounted data and models



Testing, Architecture & Project Quality

35 / 35

Automated Tests Passed

1 non-blocking warning observed during the run

Engineering Quality Highlights

- Modular source layout: src/data, ml, database, talk_to_data, utils
- Saved, versioned model artifacts (pipeline + metadata + comparison)
- SQLite persistence for applicant and prediction data
- Centralized, environment-based configuration (.env)
- Fully Dockerized deployment with health checks



Final Outcome, Limitations & Future Enhancements



Delivered

- EDA with business insights
- ML model comparison & selection
- SHAP explainability
- Talk-to-Data (NL-to-SQL)
- Streamlit multi-section UI
- Dockerized deployment



Known Limitations

- Trained on 50,000-row local sample
- Default provider is deterministic mock
- Risk thresholds not cost-calibrated
- No production security hardening
- No fairness/bias audit yet



Future Enhancements

- Full-dataset training (higher memory)
- Hyperparameter optimization
- Additional explainability (LIME)
- Business-cost threshold calibration
- Real-LLM-only mode, cloud deploy
- CI/CD and monitoring

Thank You

Questions?

AI-Powered Credit Risk Intelligence Platform

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